DAT246 – ABC & Case Study Research Summary Sheet

(Detailed 1■Page Reference, English + ■■■■■)

1■■ The ABC Framework (Stol et al., 2018)
Axes: Actors (■■■), Behavior (■■), Context (■■).
• Actors = people, organizations; Behavior = coordination / performance; Context = lab, industry, classroom.
Goal: Position SE research by **obtrusiveness (■■■■)** and **generalizability (■■■■)**.

Eight research strategies:
1 Field Study (■■■■) – observe behavior in natural context.
2 Field Experiment (■■■■) – manipulate variables in field.
3 Experimental Simulation (■■■■) – simulate real world under control.
4 Laboratory Experiment (■■■■■) – high control but low realism.
5 Judgment Study (■■■■) – experts assess artifacts / solutions.
6 Sample Study (■■■■) – survey / quantitative generalization.
7 Formal Theory (■■■■) – conceptual / analytical models.
8 Computer Simulation (■■■■■) – simulate software processes.

Tension: Cannot maximize realism, precision, and generalizability simultaneously.
Exam phrase: “Seek generalizability over responses, not over the population of actors.”
2■■ Case Study Research (Yin 2003; Runeson & Höst 2009)
Definition: Empirical inquiry investigating a **real■life phenomenon (■■■■■■)** using multiple sources of evidence.
Purpose: Explore (■■), Describe (■■), Explain (■■), Improve (■■).
Types: (1) Theory Building (Eisenhardt 1989, ■■■), (2) Theory Testing (Yin 2003, ■■■), (3) Problem Solving (Van de Ven 2007, ■■■).

5 Major Steps:
1 Design – define objectives, case selection, theory frame, RQs.
2 Preparation – create protocol (■■■■), plan data collection.
3 Data Collection – interviews, observations, archives, metrics.
4 Analysis – qualitative + quantitative triangulation (■■■■).
5 Reporting – transparent link findings ↔ data.

Uses: Motivation (falsify theories), Inspiration (theory building), Illustration (concrete example).
Sampling: statistical (■■■■■) vs analytical (■■■■ / replication logic).
3■■ Validity & Data Types (■■■■■■■)
Construct ■■■■Measure what we intend to measure. **Internal** ■■■■Control confounding / causal factors. **External** ■■■■Generalize to other contexts / actors. **Reliability** ■■Repeatability by other researchers.
Triangulation (■■■■): Combine data sources to increase validity.
Data degrees: (1) Direct (interviews), (2) Indirect (video, logs), (3) Independent (artifact analysis).
4■■ Common Misunderstandings (Flyvbjerg 2006)
1 Practical knowledge < Theoretical → ■ Both important.
2 Cannot generalize from 1 case → ■ Analytical generalization possible.
3 Case studies only generate hypotheses → ■ Also test / refine theory.
4 Biased toward verification → ■ Protocols + triangulation control bias.
5 Hard to summarize → ■ Rich context adds depth, not weakness.

Exam Tip: Focus on context uniqueness and **analytical generalization (■■■■)**, not statistical generalization.